1 Gain law

($e_\theta = \theta - \hat{\theta}$; coefficients at the estimate; $\bar{w}_s$ known)

$$\bar{a}(\bar{w}, b) = 1 - \frac{1}{b\,(\bar{w} - \bar{w}_s)}$$

$$\frac{\partial \bar{a}}{\partial \bar{w}} = b\big(1 - \bar{a}\big)^2, \qquad \frac{\partial^2 \bar{a}}{\partial \bar{a}\,\partial \bar{w}} = -2\,b\big(1 - \bar{a}\big), \qquad \Big(\frac{\partial \bar{a}}{\partial b}\Big)_{\bar{w},\,\bar{w}_s} = \frac{1 - \bar{a}}{b}$$

2 Substitution

$$e_{\bar{a}_w}[k] = -\,\hat{b}[k]\big(1 - \hat{\bar{a}}_w[k]\big)^2 e_{\delta\bar{w}_3}[k] + \frac{1 - \hat{\bar{a}}_w[k]}{\hat{b}[k]}\, e_b[k]$$

3 State

$$\mathbf{x}^{\text{original}} = \mathbf{x}^{\text{substituted}} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ b\,\big]^\top$$

$$\mathbf{x}^{\text{4-state}} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ b\,\big]^\top$$

$$\hat{\bar{w}}[k] = \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k], \qquad \hat{\bar{a}}_w[k] = 1 - \frac{1}{\hat{b}[k]\big(\hat{\bar{w}}[k] - \bar{w}_s\big)}$$

4 Error dynamics

Original

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 & \begin{array}{c} 1+\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 F_{dw} \\ -2\hat{b}\big(1-\hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d+(1-\lambda_c)\,\delta\hat{\bar{w}}_3\big) \end{array} & \begin{array}{c} \big(1-\hat{\bar{a}}_w\big)^2 \\ \times\big(\Delta\bar{w}_d+(1-\lambda_c)\,\delta\hat{\bar{w}}_3\big) \end{array} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Substituted

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c + F_{dw}\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 & 0 & -F_{dw}\dfrac{1-\hat{\bar{a}}_w}{\hat{b}} \\ 0 & 0 & \begin{array}{c} (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 \\ -F_{dw}\,\hat{b}^2\big(1-\hat{\bar{a}}_w\big)^4 \end{array} & 1-2\hat{b}\big(1-\hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d+(1-\lambda_c)\,\delta\hat{\bar{w}}_3\big) & \begin{array}{c} \big(1-\hat{\bar{a}}_w\big)^2\big(\Delta\bar{w}_d+(1-\lambda_c)\,\delta\hat{\bar{w}}_3\big) \\ +F_{dw}\big(1-\hat{\bar{a}}_w\big)^3 \end{array} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

4-state

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c + F_{dw}\,\hat{b}\big(1-\hat{\hat{a}}_w\big)^2 & -F_{dw}\dfrac{1-\hat{\hat{a}}_w}{\hat{b}} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

5 Measurement

$\big(\nabla_d \bar{w}_d[k] = \bar{w}_d[k] - \bar{w}_d[k-d];\;\; \hat{\bar{a}}_d[k] = 1 - 1/\big[\hat{b}[k]\big(\hat{\bar{w}}[k] - \nabla_d \bar{w}_d[k] - \bar{w}_s\big)\big]\big)$

Original

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + 2\hat{b}\big(1 - \hat{\bar{a}}_w\big)\nabla_d \bar{w}_d & -\big(1 - \hat{\bar{a}}_w\big)^2 \nabla_d \bar{w}_d \end{bmatrix}$$

Substituted

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\Big(1 + 2\hat{b}\big(1 - \hat{\bar{a}}_w\big)\nabla_d \bar{w}_d\Big) \times \hat{b}\big(1 - \hat{\bar{a}}_w\big)^2 & 0 & \Big(1 + 2\hat{b}\big(1 - \hat{\bar{a}}_w\big)\nabla_d \bar{w}_d\Big)\dfrac{1 - \hat{\bar{a}}_w}{\hat{b}} - \big(1 - \hat{\bar{a}}_w\big)^2 \nabla_d \bar{w}_d \end{bmatrix}$$

4-state

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -\hat{b}\big(1 - \hat{\bar{a}}_d\big)^2 & \dfrac{1 - \hat{\bar{a}}_d}{\hat{b}} \end{bmatrix}$$

6 Observability

($\mathbf{e}[k]$ the error vector of the version at hand, n its length; $\mathbf{n}[k] = [\, n_w[k],\ n_a[k]\,]^\top$
$\Phi(k, k_0) := F_e[k-1]\, F_e[k-2] \cdots F_e[k_0]$, $\Phi(k_0, k_0) = I$
N window length
$\mathbf{Y} := [\, \mathbf{e}_y[k_0];\ \ldots;\ \mathbf{e}_y[k_0+N]\,]$, $\mathbf{N} := [\, \mathbf{n}[k_0];\ \ldots;\ \mathbf{n}[k_0+N]\,]$
$X_{:,j}$ column j of X)

$$\mathbf{e}[k] = \Phi(k, k_0)\, \mathbf{e}[k_0]$$
$$\mathbf{e}_y[k] = H[k]\, \Phi(k, k_0)\, \mathbf{e}[k_0] + \mathbf{n}[k]$$
$$\mathbf{Y} = O(k_0, N)\, \mathbf{e}[k_0] + \mathbf{N}, \qquad O(k_0, N) := \begin{bmatrix} H[k_0]\, \Phi(k_0, k_0) \\ H[k_0+1]\, \Phi(k_0+1, k_0) \\ \vdots \\ H[k_0+N]\, \Phi(k_0+N, k_0) \end{bmatrix}$$

$$R_w := \operatorname{diag}\big(R[k_0],\ \ldots,\ R[k_0+N]\big), \qquad O_w(k_0, N) := R_w^{-1/2}\, O(k_0, N)$$
$$\mathbf{Y}_w := R_w^{-1/2}\, \mathbf{Y} = O_w(k_0, N)\, \mathbf{e}[k_0] + \mathbf{N}_w, \qquad \mathbb{E}\big[\mathbf{N}_w \mathbf{N}_w^\top\big] = I$$

$$O_w(k_0, N)\, \mathbf{v}_i = \sigma_i\, \mathbf{u}_i, \qquad i = 1, \ldots, n, \qquad \|\mathbf{v}_i\| = \|\mathbf{u}_i\| = 1$$
$$\sigma_1 \geq \cdots \geq \sigma_n \geq 0$$
$$\sigma_1 = \max_{\|\mathbf{v}\|=1} \|O_w(k_0, N)\, \mathbf{v}\|, \qquad \sigma_n = \min_{\|\mathbf{v}\|=1} \|O_w(k_0, N)\, \mathbf{v}\|$$
$$\operatorname{rank} O_w(k_0, N) = \#\{\, i : \sigma_i > 0\,\}$$
$$F_e[k]_{:,j} = f_j[k]\, I_{:,j}, \quad H[k]_{:,j} = \mathbf{0}\ \ \forall k \ \Rightarrow\ O_w(k_0, N)_{:,j} = \mathbf{0}$$
$$\Rightarrow\ \sigma_n = 0$$

The two columns

$$\text{original:}\quad F_{e,:,\bar{a}_w} = \Big[\, 0,\ 0,\ -F_{dw},\ 1 + \hat{b}\big(1 - \hat{\bar{a}}_w\big)^2 F_{dw} - 2\hat{b}\big(1 - \hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d + (1-\lambda_c)\, \delta\hat{\bar{w}}_3\big),\ 0\,\Big]^\top$$
$$H_{:,\bar{a}_w} = \Big[\, 0,\ 1 + 2\hat{b}\big(1 - \hat{\bar{a}}_w\big)\nabla_d\bar{w}_d\,\Big]^\top$$
$$\text{substituted:}\quad F_{e,:,\bar{a}_w} = \Big[1 - 2\hat{b}\big(1 - \hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d + (1-\lambda_c)\, \delta\hat{\bar{w}}_3\big)\Big]\, I_{:,\bar{a}_w}$$
$$H_{:,\bar{a}_w} = \mathbf{0}$$

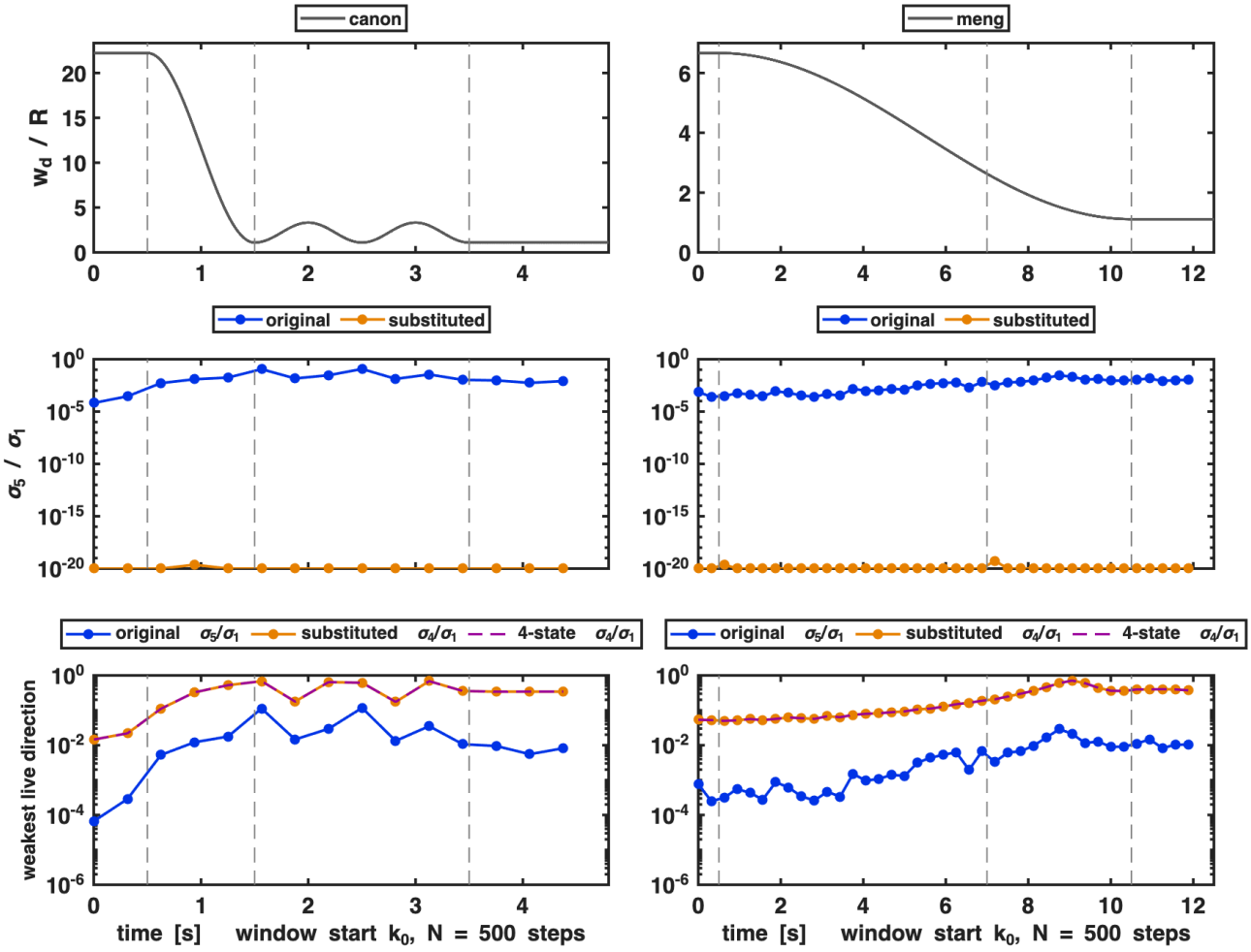
$$\text{original:}\quad F_{e,:,b} = \Big[\, 0,\ 0,\ 0,\ \big(1 - \hat{\bar{a}}_w\big)^2\big(\Delta\bar{w}_d + (1-\lambda_c)\, \delta\hat{\bar{w}}_3\big),\ 1\,\Big]^\top, \quad H_{:,b} = \Big[\, 0,\ -\big(1 - \hat{\bar{a}}_w\big)^2\nabla_d\bar{w}_d\,\Big]^\top$$

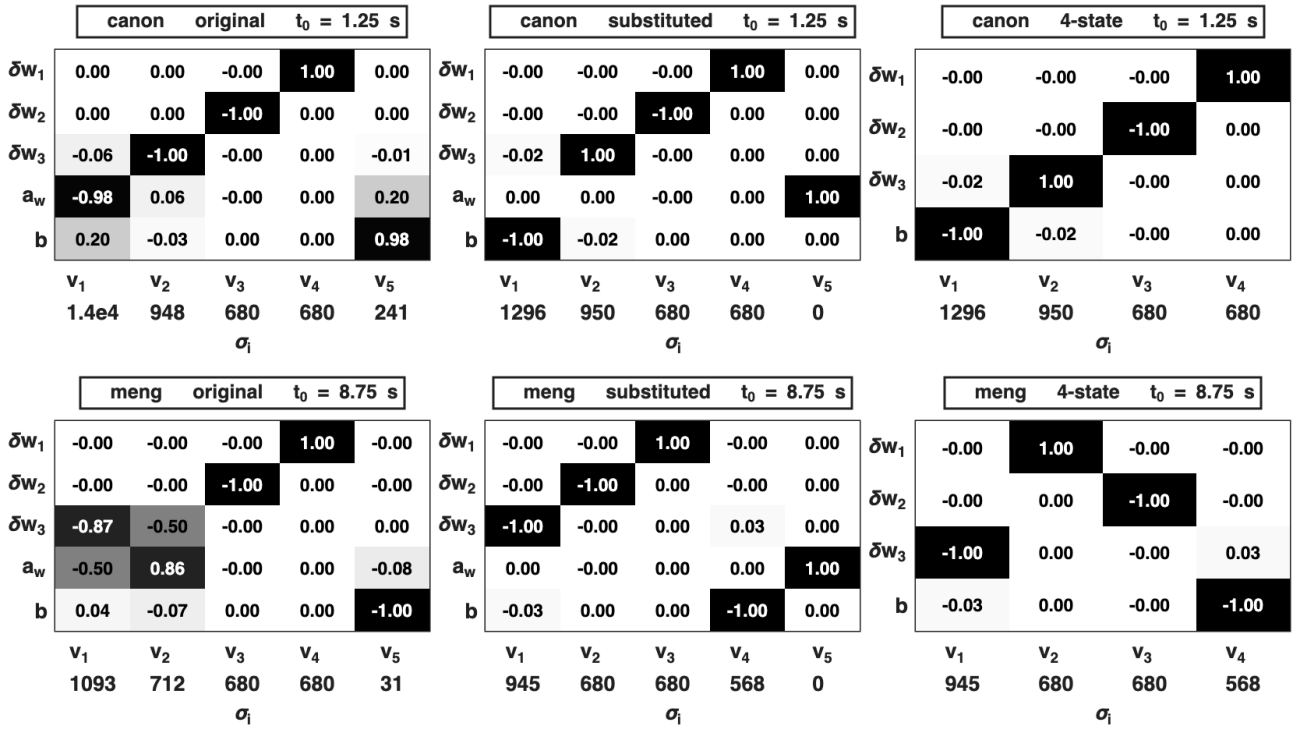
$$\text{4-state:}\quad F_{e,:,b} = \Big[\, 0,\ 0,\ -F_{dw}\frac{1 - \hat{\bar{a}}_w}{\hat{b}},\ 1\,\Big]^\top, \qquad H_{:,b} = \Big[\, 0,\ \frac{1 - \hat{\bar{a}}_d}{\hat{b}}\,\Big]^\top$$

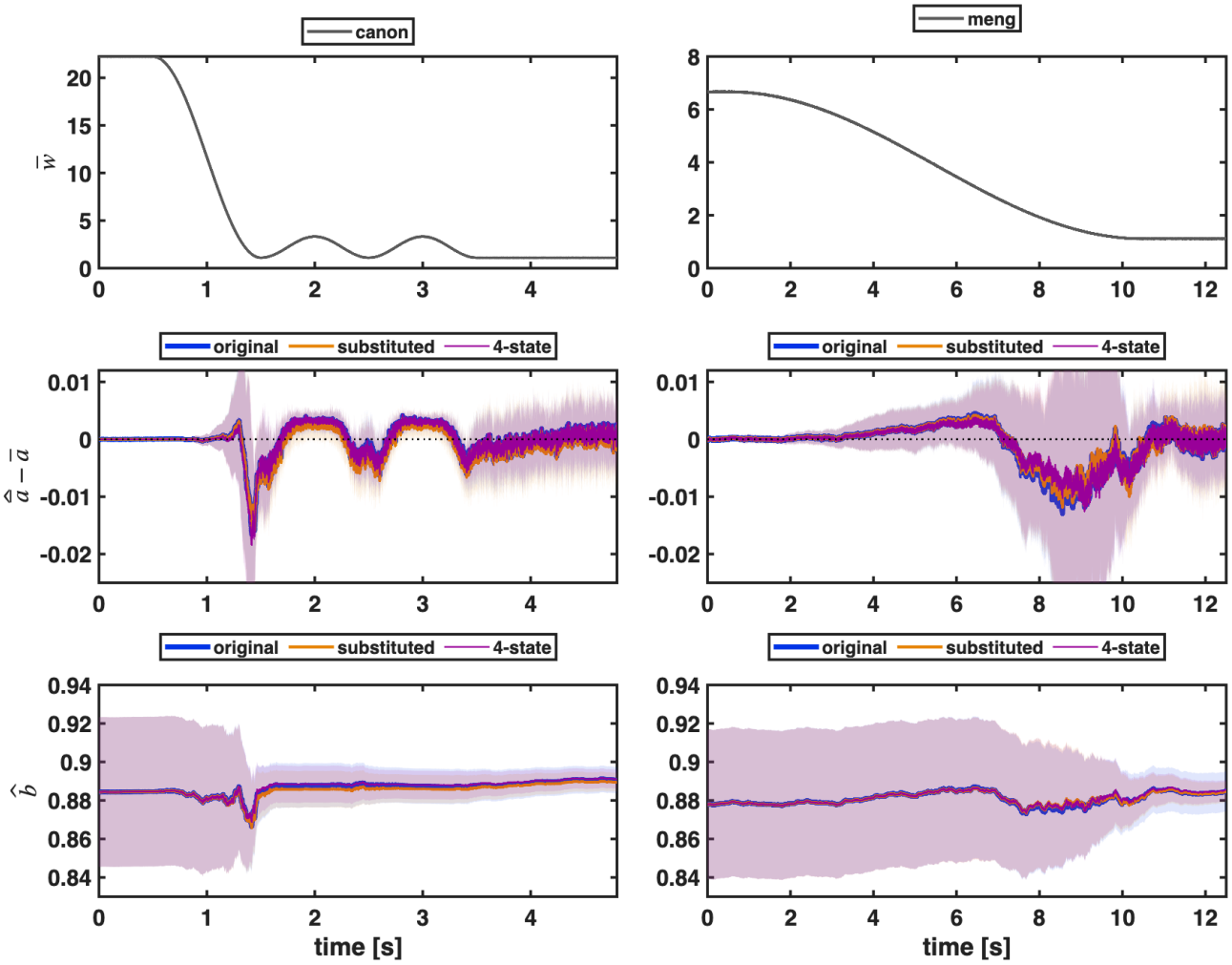
$$\Delta\bar{w}_d[k] = 0 \;\; \forall k \quad \Longrightarrow \quad F_{dw} \to 0, \qquad \nabla_d \bar{w}_d = 0$$
$$\Longrightarrow \quad H^{\text{original}}_{:,b} = \mathbf{0}, \qquad H^{\text{4-state}}_{:,b} = \left[\, 0, \;\; \frac{1 - \hat{\bar{a}}_w}{\hat{b}} \,\right]^\top \neq \mathbf{0}$$

Ratios

$$\frac{\sigma_5}{\sigma_1} \;\; (n = 5), \qquad \frac{\sigma_4}{\sigma_1} \;\; (n = 4), \qquad \frac{\sigma_n}{\sigma_1} = 0 \iff \exists\, j : \; O_w(k_0, N)_{:,j} = \mathbf{0}$$


canon
meng
w_d / R
original
substituted
σ_5 / σ_1
original σ_5/σ_1
substituted σ_4/σ_1
4-state σ_4/σ_1
weakest live direction
time [s] window start k_0, N = 500 steps


canon original t_0 = 1.25 s
δw_1 0.00 0.00 -0.00 1.00 0.00
δw_2 0.00 0.00 -1.00 0.00 0.00
δw_3 -0.06 -1.00 -0.00 0.00 -0.01
a_w -0.98 0.06 -0.00 0.00 0.20
b 0.20 -0.03 0.00 0.00 0.98
v_1 v_2 v_3 v_4 v_5
1.4e4 948 680 680 241
σ_i
canon substituted t_0 = 1.25 s
δw_1 -0.00 -0.00 -0.00 1.00 0.00
δw_2 -0.00 -0.00 -1.00 0.00 0.00
δw_3 -0.02 1.00 -0.00 0.00 0.00
a_w 0.00 0.00 -0.00 0.00 1.00
b -1.00 -0.02 0.00 0.00 0.00
v_1 v_2 v_3 v_4 v_5
1296 950 680 680 0
σ_i
canon 4-state t_0 = 1.25 s
δw_1 -0.00 -0.00 -0.00 1.00
δw_2 -0.00 -0.00 -1.00 0.00
δw_3 -0.02 1.00 -0.00 0.00
b -1.00 -0.02 -0.00 0.00
v_1 v_2 v_3 v_4
1296 950 680 680
σ_i
meng original t_0 = 8.75 s
δw_1 -0.00 -0.00 -0.00 1.00 -0.00
δw_2 -0.00 -0.00 -1.00 0.00 -0.00
δw_3 -0.87 -0.50 -0.00 0.00 0.00
a_w -0.50 0.86 -0.00 0.00 -0.08
b 0.04 -0.07 0.00 0.00 -1.00
v_1 v_2 v_3 v_4 v_5
1093 712 680 680 31
σ_i
meng substituted t_0 = 8.75 s
δw_1 -0.00 -0.00 1.00 -0.00 0.00
δw_2 -0.00 -1.00 0.00 -0.00 0.00
δw_3 -1.00 -0.00 0.00 0.03 0.00
a_w 0.00 -0.00 0.00 0.00 1.00
b -0.03 0.00 0.00 -1.00 0.00
v_1 v_2 v_3 v_4 v_5
945 680 680 568 0
σ_i
meng 4-state t_0 = 8.75 s
δw_1 -0.00 1.00 -0.00 -0.00
δw_2 -0.00 0.00 -1.00 -0.00
δw_3 -1.00 0.00 -0.00 0.03
b -0.03 0.00 -0.00 -1.00
v_1 v_2 v_3 v_4
945 680 680 568
σ_i


canon
meng
$\bar{w}$
original
substituted
4-state
$\hat{a}-\bar{a}$
$\hat{b}$
time [s]